

4.2 Graphics Cards

“Graphics cards” refers to expansion cards that contain additional hardware to ease the computational load on the CPU when dealing with graphics data. Graphics cards contains two important components, the GPU and the Frame Buffer.

4.2.1 AGP

Accelerated Graphics Port is a type of expansion slot dedicated for the use of graphics cards. The maximum bandwidth offered is 2133 Megabytes per second.

4.2.2 Graphics Aperture

Some part of the system RAM can be reserved for use by the graphics card to store data that exceeds its Frame Buffer. This is called the Graphics Aperture, or just Aperture. This value is set in the system BIOS.

4.2.3 Alpha Blending

This refers to the process by which information in the alpha channel is included with the colour characteristics of a pixel. The alpha channel carries information related to the transparency of the pixel, and is an 8-bit code capable of displaying 256 levels of transparency. A 32-bit colour scheme includes 8 bits of information for the Red, Green, Blue, and Alpha channels.

4.2.4 Anisotropic Filtering (AF)

The most advanced form of texture filtering (explained later), it works in conjunction with bilinear and trilinear filtering to correct artefacts from a larger perspective. Since the user’s view reveals object surfaces in different shapes—for example, a rectangular ceiling appears as a trapezoid—texture filtering has to ensure an even transition of the texture laid on this shape, and not the original shape of the object. Anisotropic Filtering ensures that bilinear and trilinear filtering achieves the correct transition effects by changing the sampling pattern based on the user’s perspective. AF can also be denoted by the number of samples taken, like AF 8X.